

MULTI-SEGMENTAL FLEXION



Test Objective for the Multi-Segmental Flexion:

The Multi-Segmental Flexion assessment tests for normal flexion in the hips and spine.

How to Perform the Multi-Segmental Flexion:

The patient assumes the starting position by standing erect with feet together and toes pointing forward. They then bend forward from hips reaching down to touch the ends of their fingers to the tips of their toes without bending their knees. Direct the patient to look down towards toes during the movement.

Functional Criteria:

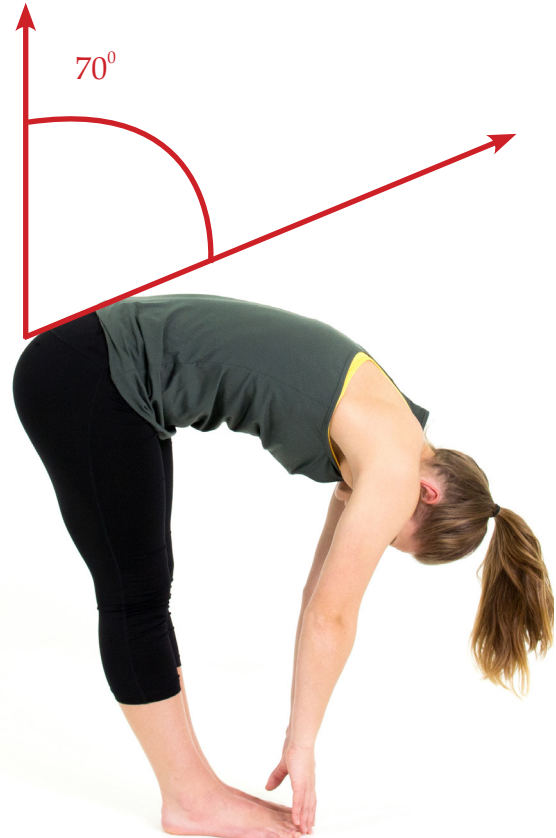
- Touches toes
- Posterior weight shift
- Uniform spine curve
- Sacral Angle of at least 70 degrees
- Without excessive effort and/or loss of motor control

Common Compensations:

- Knee bend

Tips for Testing:

- Observe from the side view.
- Foot position should remain unchanged throughout the movement.
- Knees should remain straight.
- If patient touches the floor in front of their toes, instruct them to only touch to their toes.
- Cervical spine should not be extended
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



MULTI-SEGMENTAL FLEXION BREAKOUT



Multi-Segmental Flexion Breakout Rationale

Weight Bearing or Non-Weight Bearing Dysfunction

We can change the stability requirements of the lower extremities by having the patient sit down, legs straight, for the Long Sitting Test. With the client sitting tall, have them bend forward and try to touch their toes, keeping their knees straight. Remember they must demonstrate a full 80° sacral angle, a uniform spine curve and touch their toes to call this movement functional.

If the Long Sitting Test is functional and non-painful, then assume this is a weight-bearing hip flexion pattern stability/motor control dysfunction (SMCD). If the Long Sitting Test is dysfunctional and/or painful, then we must assume there is a problem with hip flexion and/or spine flexion. We need to check both parts individually and try to identify where the dysfunction and/or pain is coming from. We start by checking hip flexion with the Active Straight Leg Raise Test.

Hip Flexion Problem

With the patient lying supine, have them perform an Active Straight Leg Raise. Be aware of any lumbar flexion and/or excessive posterior tilt - this is no longer isolated hip flexion. If the test is functional and non-painful, we will assume hip flexion is normal and progress to check spine flexion with the Prone Rocking Test. If the Active Straight Leg Raise Test is dysfunctional and/or painful, then we will progress to the Stabilized Active Straight Leg Raise Test.

Pelvic orientation can directly affect the spacing in the acetabulum and therefore, give the impression that there is a hip mobility problem, when in reality it may just be a SMCD. The Stabilized Active Straight Leg Raise Test engages the core to help reorient the pelvis and then allows us to retest the ASLR test. If function is restored and pain is eliminated, there is an underlying core (pelvic orientation) stability/motor control dysfunction creating the hip flexion dysfunction. If it is still dysfunctional and/or painful, we will progress to the Passive Straight Leg Raise Test.

If the Passive Straight Leg Raise Test is functional and non-painful, we have an active hip flexion SMCD. We have our diagnosis for the lower body, but we still need to check spine flexion with the Prone Rocking Test. If the Passive Straight Leg Raise is dysfunctional and/or painful, then we either have a posterior chain mobility dysfunction or a hip flexion mobility dysfunction. To check the hip joint, we progress to the Supine Knees to Chest Test.

In the Supine Knees to Chest Test, make sure the patient keeps their knees together. Have them pull their knees to their chest while holding their thighs, without their lumbar spine flexing. If this is functional and non-painful, we know that the hip joint is normal for flexion. Since we were able to clear the hip joint for flexion, our diagnosis would be a posterior chain mobility dysfunction or if the Passive Straight Leg Raise Test was functional and painful it could also be an active hip flexion SMCD.

If the Supine Knees to Chest Test was dysfunctional then assume we have a hip flexion mobility dysfunction. This rules in the hip flexion mobility dysfunction, but remember, they can also have a posterior chain MD.

MULTI-SEGMENTAL FLEXION BREAKOUT



If Supine Knees to Chest Test was painful(FP or DP), note where the pain is and come back and examine the region after completing the entire SFMA.

No matter your finding for this test, we need to proceed and check spine flexion with the Prone Rocking Test.

Spine Flexion Problem

The last part we need to assess is the spine flexion component using the Prone Rocking Test. This is a great test to check unloaded spine flexion. With the patient in the quadruped position, have them sit back onto their heels as far as they comfortably can and try to flex their spine as much as possible. Their arms should rest comfortably at their sides. You should see a uniform spinal curve and full flexion of the spine.

If the Prone Rocking Test is functional and non-painful, and you observed a non-uniform spine curvature in any previous test, there is a weight-bearing spine flexion SMCD. If the Prone Rocking Test is functional and non-painful, and they demonstrated a uniform spine curvature in previous tests, consider spine flexion normal. If spine flexion is normal and there were no previous findings, consider a possible plantarflexion dysfunction affecting Multi-Segmental Flexion.

If the Prone Rocking Test was dysfunctional and non-painful, consider this a spine flexion mobility dysfunction.

If the Prone Rocking Test was painful(FP or DP), note where the pain is and come back and examine the region after completing the entire SFMA.

MULTI-SEGMENTAL FLEXION BREAKOUT



Selective Breakout Tests for Multi-Segmental Flexion:



Long Sitting Test



Active Straight
Leg Raise
(ASLR) Test



Stabilized Active
Straight Leg Raise
(SASLR) Test



Passive Straight Leg
Raise (PSLR) Test

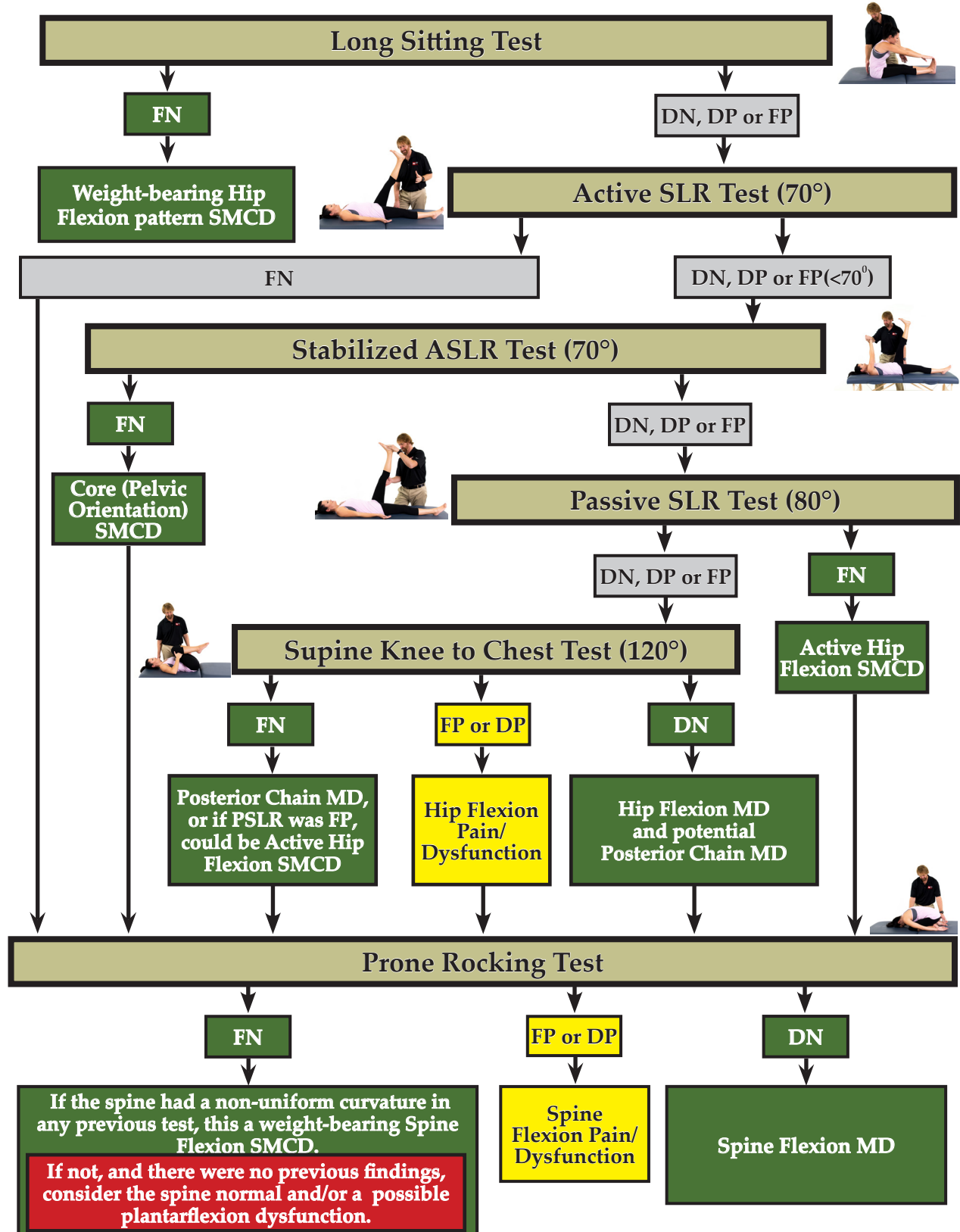


Supine Knee to Chest
Holding Thighs



Prone Rocking Test

MULTI-SEGMENTAL FLEXION BREAKOUT



MULTI-SEGMENTAL FLEXION BREAKOUT



Test Objective for the Long Sitting Test:

To test the forward flexion movement while unloading the lower body.

How to Perform the Long Sitting Test:

The patient will be in a long sitting position with their legs extended, knees together, and ankles neutral. Instruct the patient to bend forward to touch their toes.

Functional Criteria:

A normal Long Sitting Test includes a uniform spinal curve, sacral angle of 80 degrees or greater, and ability to touch their toes with their knees straight.

Additional Information:

Sacral angle is measured from the horizontal up - whereas the top tier test was measured from the vertical down.

Possible findings:

- FN: Touches toes with a sacral angle of 80 degrees or greater, uniform spinal curve, no pain
 - » Indicates a weight-bearing hip flexion pattern stability/motor control dysfunction.
- FP: Touches toes with a sacral angle of 80 degrees or greater, uniform spine curve, pain
 - » Proceed to Active Straight Leg Raise Test.
- DN (DP if pain is present): Touches toes with less than 80 degrees of sacral angle and/or non-uniform spinal curve
 - » Proceed to Active Straight Leg Raise Test.
- DN (DP if pain is present): Can or cannot touch toes and has at least 80 degrees of sacral angle and/or non-uniform spinal curve
 - » Proceed to Active Straight Leg Raise Test.
- DN (DP if pain is present): Cannot touch toes, less than 80 degrees of sacral angle and/or non-uniform spinal curve
 - » Proceed to Active Straight Leg Raise Test.



MULTI-SEGMENTAL FLEXION BREAKOUT



Test Objective for the Active Straight Leg Raise Test:

To assess the ability to actively flex the hip with a straight knee in an unloaded posture.

How to Perform the Active Straight Leg Raise Test:

Have the patient lie supine with arms down by the sides, palms facing up, and the head flat on the floor or examination table. Both ankles should be in a neutral position, the soles of the feet perpendicular to the floor or table. Instruct the patient to lift the test limb while maintaining the initial ankle position.

During the test, the opposite knee should remain in contact with the table. The toes should remain pointed upward in the neutral-limb position, and the head should remain flat on the floor or table.

Once the patient achieves the end range, measure the amount of hip flexion as compared with the leg on the ground. Test bilaterally.

Functional Criteria:

Using the femur as the reference line, normal Active Straight Leg Raise is 70 degrees or greater with the sacrum remaining in contact with the testing surface.

Additional Information:

This assessment helps identify the restriction as a symmetrical or asymmetrical dysfunction in a non-weight-bearing position. When testing, be aware of any lumbar flexion and/or posterior tilt of the pelvis.

Possible Findings:

- FN: 70 degrees or greater of active straight leg hip flexion, no pain
 - » Consider lower body flexion normal for Multi-Segmental Flexion. Proceed to the Prone Rocking Test.
- FP: 70 degrees or greater of active straight leg hip flexion, pain
 - » Proceed to the Stabilized Straight Leg Raise Test.
- DN: Less than 70 degrees of active straight leg hip flexion, no pain
 - » Proceed to the Stabilized Straight Leg Raise Test.
- DP: Less than 70 degrees of active straight leg hip flexion, pain
 - » Proceed to the Stabilized Straight Leg Raise Test.



MULTI-SEGMENTAL FLEXION BREAKOUT



Test Objective for the Stabilized Active Straight Leg Raise Test:

To assess for an underlying SMCD of the core that may be limiting hip mobility. Pelvic orientation can directly affect the spacing in the acetabulum and therefore give the impression that there is a hip mobility problem, when in reality it may just be a stability/motor control problem.

How to Perform the Stabilized Active Straight Leg Raise Test:

Have the patient lie supine, legs in hook-lying position, arms raised toward ceiling and the head flat on the table. Now, have the patient push against resistance with their hands towards the floor to activate the core and create a temporary neutral tilt of the pelvis. Once stabilized, the patient will lower one leg at a time into the original ASLR testing position. Once both legs are in original ASLR testing position, the patient should immediately lift the testing leg into an ASLR. The patient will continue to push against resistance throughout the entire test.

During the test, the opposite knee should remain in contact with the table. The toes should remain pointed upward in the neutral-limb position, and the head should remain flat on the floor or table. Once the patient achieves the end range, measure the amount of hip flexion as compared with the leg on the ground. Look for any change in the restriction or amount of pain. Repeat on the other side.

Functional Criteria:

Normal motion would be an ASLR of 70 degrees or greater with sacrum remaining in contact with testing surface - and/or if there was pain present, it is now eliminated.

Additional Information:

You can activate the core by resisting manually with your hands against the patient's hands or use some form of elastic tubing to create the resistance. When testing, be aware of any lumbar flexion or excessive posterior tilting.

Possible Findings:

- FN: 70 degrees or greater of active straight leg hip flexion, no pain
 - » Indicates a Core (Pelvic Orientation) stability/motor control dysfunction. Proceed to the Prone Rocking Test.
- FP: 70 degrees or greater of active straight leg hip flexion, pain
 - » Proceed to the Passive Straight Leg Raise Test.
- DN: Less than 70 degrees of active straight leg hip flexion, no pain
 - » Proceed to the Passive Straight Leg Raise Test.
- DP: Less than 70 degrees of active straight leg hip flexion, pain
 - » Proceed to the Passive Straight Leg Raise Test.

Step 1:



Step 2:



Step 3:



MULTI-SEGMENTAL FLEXION BREAKOUT



Test Objective for the Passive Straight Leg Raise Test:

To assess the ability to passively flex the hip with a straight knee in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Straight Leg Raise Test:

Instruct the patient to lie supine with the arms down by the sides. Keeping the right leg extended and flat on the ground or table, passively lift the left leg slowly. Ankles should remain in neutral position. Go until first resistance is felt without any knee bend or pelvic shift. Measure the amount of hip flexion as compared with the leg on the ground. Test bilaterally.

Functional Criteria:

Normal Passive Straight Leg Raise is 80 degrees or greater with the sacrum remaining in contact with the testing surface.

Additional Information:

When testing, be aware of movement in the down leg as well as any lumbar flexion or excessive posterior tilting.

Possible Findings:

- FN: 80 degrees or greater of passive straight leg hip flexion, no pain
 - » Indicates an active hip flexion stability/motor control dysfunction. Proceed to the Prone Rocking Test.
- FP: 80 degrees or greater of passive straight leg hip flexion, pain
 - » Proceed to the Supine Knee to Chest test.
- DN: Less than 80 degrees of passive straight leg hip flexion, no pain
 - » Proceed to the Supine Knee to Chest Test.
- DP: Less than 80 degrees of passive straight leg hip flexion, pain
 - » Proceed to the Supine Knee to Chest Test.



MULTI-SEGMENTAL FLEXION BREAKOUT



Test Objective for the Supine Knee to Chest Holding Thighs Test:

To assess the mobility of the hips into hip flexion in an unloaded posture.

How to Perform the Supine Knee to Chest Holding Thighs Test:

Have the patient lie supine and bring both knees up, keeping the knees together. The patient should then grab the back of the thighs and pull the thighs to the chest.

Functional Criteria:

Normal hip flexion with knees bent is 120 degrees with the sacrum remaining in contact with floor/table. This is commonly when the thighs lay against the lower rib cage.

Additional Information:

This move will help differentiate between a hip mobility dysfunction and a straight leg hip flexion (posterior chain) mobility dysfunction.

Possible Findings:

- FN: 120 degrees of hip flexion, no pain
 - » Indicates a posterior chain mobility dysfunction, or if PSLR was FP, an active hip flexion stability/motor control dysfunction. Proceed to Prone Rocking Test.
- FP: 120 degrees of hip flexion, pain
 - » Indicates pain with hip flexion - complete local biomechanical exam. Proceed to Prone Rocking Test.
- DN: Less than 120 degrees of hip flexion, no pain
 - » Indicates a hip mobility dysfunction for hip flexion - complete local biomechanical exam. Proceed to Prone Rocking Test.
- DP: Less than 120 degrees of hip flexion, pain
 - » Indicates a painful hip flexion mobility dysfunction - complete local biomechanical exam. Proceed to the Prone Rocking Test.



MULTI-SEGMENTAL FLEXION BREAKOUT



Test Objective for the Prone Rocking Test:

To assess spinal flexion in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Prone Rocking Test:

Instruct the patient to start in the quadruped, all-fours position. Have the patient lower the buttocks onto the heels—rocking back—toward full hip flexion. Their arms should rest comfortably by their sides.

Functional Criteria:

The spine should have a uniform forward curvature and the lower rib cage should easily press into the thighs at the end of the rock-back.

Additional Information:

The knee can be a complicating factor. It is appropriate in this situation to have the patient sit in a chair and bend forward to assess the curvature of their spine.

Possible Findings:

- FN: Uniform spinal curve, no pain
 - » If the spine had a non-uniform curvature in any previous test, this indicates a weight-bearing spine flexion stability/motor control dysfunction. If not, and there were no previous findings, consider the spine normal and/or a possible plantarflexion dysfunction.
- FP: Uniform spinal curve, pain
 - » Indicates pain with spine flexion - complete local biomechanical exam.
- DN: Non-uniform spinal curve, no pain
 - » Indicates a spine flexion mobility dysfunction - complete local biomechanical exam.
- DP: Non-uniform spinal curve, pain
 - » Indicates a painful spine flexion mobility dysfunction - complete local biomechanical exam.



MULTI-SEGMENTAL EXTENSION BREAKOUTS



Multi-Segmental Extension Breakout Rationale

Multi-Segmental Extension Breakout is divided into two separate flowcharts - Spine/Upper Body Extension and Lower Body Extension. We always start with the Spine/Upper Body Extension flowchart and it will tell us where to go from there.

Spine/Upper Body Extension Flowchart

We start by combining the first two steps of our logic, removing body parts and changing the stability requirements, to assess the spine. Have the patient lie prone and perform a press-up. In this posture, we can assess how the patient's non-weight-bearing extension pattern compares to a full weight-bearing extension pattern.

If the press-up is functional and non-painful, consider spine extension normal, but we do also need to check for a shoulder flexion problem with the Active Prone Shoulder Girdle Flexion Test. We can use active versus passive testing, if needed, to differentiate between mobility and stability/motor control dysfunction before moving on to check the Lower Body Extension flowchart.

Thorax Extension/Rotation Problem

If the press-up is dysfunctional and/or painful, we must further breakout spine extension. To differentiate between the thorax and lumbar regions, we use the Active Lumbar Locked (IR) Extension/Rotation Test. The lumbar lock position allows better isolation of the thorax and reduces the contribution from the lumbar spine. Placing the arm behind the back in an internally rotated position allows a reduction in scapular involvement and minimizes anterior shoulder and chest dysfunction.

If thorax mobility is functional and non-painful bilaterally, then assume the thorax is meeting minimum active movement mobility and motor control and continue to evaluate the lumbar spine with the Prone on Elbow Test. If there is dysfunction and/or pain (either bilateral or unilateral) with Active Lumbar Locked Extension/Rotation Test, we retest the motion passively to differentiate between a mobility or stability/motor control dysfunction.

If the Passive Lumbar Locked (IR) Extension/Rotation motion is dysfunctional, with or without pain, there is a thorax extension/rotation mobility dysfunction. At this point, we have ruled in a thorax mobility dysfunction - we have not ruled out a lumbar dysfunction. A clinical pearl within the SFMA is that a Thorax Mobility Dysfunction finding is a priority over Lumbar Dysfunction finding based on our mobility/stability model. Therefore, we suggest treating the thorax mobility dysfunction prior to assessing the lumbar region. There may also be a shoulder girdle dysfunction, so it is appropriate to continue on to the Active Prone Shoulder Girdle Test.

If the Passive Lumbar Locked Extension/Rotation test is now functional, with or without pain, there is a thorax extension/rotation SMCD. If mobility of the thorax is cleared bilaterally passively, with or without pain, we must continue to assess the lumbar by proceeding to Prone on Elbow Test.

MULTI-SEGMENTAL EXTENSION BREAKOUTS



Lumbar Spine Extension Problem

To assess the lumbar region, we use the Prone on Elbow Test. In this position, the thorax is already extended, so the stress will now be placed on the lumbar region. If extension from this position creates pain, stop and note where the pain is and come back and examine the region after completing the entire SFMA.

If lumbar extension/rotation is functional and non-painful bilaterally, we must first go back and see what happened with the thorax. If the thorax was an SMCD, then we can assume the lumbar is normal. If the thorax was normal, then we must assume there is a weight-bearing spine extension SMCD or an anterior torso mobility dysfunction.

If there is dysfunction and/or pain with lumbar extension/rotation, we must proceed to passive testing to differentiate between mobility or stability/motor control dysfunction. If the Passive Prone on Elbow Test is functional and non-painful, assume the patient has a weight bearing spine extension SMCD.

If the Passive Prone on Elbow Test is dysfunctional and non-painful, then assume the patient has a lumbar extension/rotation mobility dysfunction.

If the Passive Prone on Elbow Test creates pain, stop and note where the pain is and come back and examine the region after completing the entire SFMA.

No matter your finding in the Prone on Elbow Test, we proceed to check for a shoulder dysfunction with the Active Prone Shoulder Girdle Flexion Test.

Shoulder Flexion Problem

Once the spine has been assessed, we can continue to look at shoulder flexion using the Active Prone Shoulder Girdle Flexion Test. If shoulder girdle flexion is functional and non-painful, consider shoulder girdle flexion to be normal. If there was dysfunction or pain with Active Prone Shoulder Girdle Flexion Test, we retest the motion passively to differentiate between a mobility or stability/motor control dysfunction.

If the Passive Prone Shoulder Girdle Flexion Test is functional and non-painful, this indicates a shoulder flexion stability/motor control dysfunction. If the Passive Prone Shoulder Girdle Flexion Test is dysfunctional and non-painful, then assume the patient has a shoulder girdle flexion mobility dysfunction.

If the Passive Prone Shoulder Girdle Flexion Test is painful, stop and note where the pain is and come back and examine the region after completing the entire SFMA.

This concludes our findings for the Spine and Upper Body components of Multi-Segmental Extension - we must move to the Lower Body Extension Flowchart.

MULTI-SEGMENTAL EXTENSION BREAKOUTS



Lower-Body Extension Flowchart

Hip Mobility Dysfunction

Next, let's walk through the Lower-Body Extension Breakout. Normal hip extension is at least 10 degrees, but we will first determine if the patient has a minimum of 0 degrees. We assume if they can't show us 0 degrees of hip extension, then they cannot show us 10 degrees. We will use 2 orthopaedic tests to assess for 0 degrees of hip extension.

To examine the mobility of the hip/SI joint, we use the FABER Test. If the FABER Test is normal, then we proceed to the Modified Thomas Test to evaluate the soft tissue around the hip. If the FABER Test is dysfunctional and/or painful, then we proceed to the Stabilized FABER Test. In this test, we will differentiate between a mobility or stability/motor control dysfunction at the hip. If by activating the core, essentially re-positioning the pelvic orientation, creates a functional and non-painful Stabilized FABER Test, the problem is a Core/Pelvic Orientation SMCD. If the stabilization does not change anything and the test is still dysfunctional, there is a Hip/SI mobility dysfunction. If there is pain with the Stabilized FABER Test, note where the pain is and come back and examine the region after completing the entire SFMA. No matter your finding with FABER Test, we must still proceed to the Modified Thomas test since muscular limitations may also be involved.

We use the Modified Thomas Test to assess the anterior and lateral tissue extensibility for hip extension. If the Modified Thomas Test is dysfunctional and non-painful (either unilateral or bilateral), there is a hip extension mobility dysfunction preventing the patient from reaching 0 degrees of hip extension. We must assess and treat this mobility dysfunction prior to assessing into 10 degrees of hip extension.

As always, if there is pain with the Modified Thomas Test, note where the pain is and come back and examine the region after completing the entire SFMA.

If the Modified Thomas Test is functional and non-painful bilaterally but the FABER Test was dysfunctional and/or painful (either unilateral or bilateral), we stop and assess the FABER Test findings. If the Modified Thomas Test is functional and non-painful bilaterally and the FABER Test was FN, we cleared hip extension to 0 degrees - we must proceed to test for 10 degrees of hip extension using the Prone Active Hip Extension Test.

Spine & Hip Weight Bearing SMCD

The Active Prone Hip Extension Test is used to evaluate into 10 degrees of hip extension. The only time this is tested is when FABER and the Modified Thomas Tests are both functional and non-painful bilaterally. The Active Prone Hip Extension Test position reduces the postural stabilization requirements for active hip extension. In this posture, we can assess how the patient's non-weight bearing spine and hip affects the hip extension pattern.

If hip extension is now functional and non-painful, and there are no signs of spine extension dysfunction, this indicates a weight-bearing hip extension SMCD. If spine extension was dysfunctional, assume hip extension is normal. There is a possibility ankle dorsiflexion could also be the only cause of dysfunction for the lower body, but that is extremely rare. We would catch that in the Arms Down Deep Squat or Single Leg Stance Pattern.

MULTI-SEGMENTAL EXTENSION BREAKOUTS



If hip extension is dysfunctional and/or painful in the Active Prone Hip Extension Test, we should then assess hip extension passively using the Passive Prone Hip Extension Test. If hip extension is now functional and non-painful with passive testing, there is a core SMCD and/or active hip extension SMCD.

If hip extension is dysfunctional and non-painful with passive testing, this indicates a hip extension mobility dysfunction.

If there is pain with the Passive Prone Hip Extension Test, note where the pain is and come back and examine the region after completing the entire SFMA.